Quantum Approximate Optimization Algorithm

Assessment of the QAOA algorithm (Max-Cut)

QUESTION PAPER

Time allowed:	3 hours	**Maximum marks:**	240
Questions:	80 (4 sections × 20)	**Answer key:**	separate Answer Book

Name of the candidate: _____ **Roll No.:** _____

General instructions

1. This question paper has **four sections** of **20 questions** each: **A** Mathematics and Graph Theory for QAOA, **B** Quantum Mechanics, **C** QAOA Basics, **D** Advanced QAOA and Implementation.
2. Every question is of the **single-correct** type: four options, exactly one correct. Question formats used: single-correct MCQ, matching lists (List-I/List-II), Assertion–Reason, Statement I/II and paragraph-based questions.
3. For every question choose **ONE** option, (A), (B), (C) or (D), and record it against the question number in the answer script provided by the invigilator.
4. Calculators are **not** allowed. Rough work may be done on separate sheets.
5. Unless a question says otherwise, use the **conventions** printed below; they apply to every section.
6. The detailed solutions are in the separate *Answer Book*; do not consult it before finishing the test.

Marking scheme (every question)

Each question has **four** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Full marks:	+3 if ONLY the correct option is chosen;
Zero marks:	0 if none of the options is chosen (question unanswered);
Negative marks:	−1 in all other cases (wrong option, or more than one option chosen).

Conventions and notation used throughout the paper

- Graph $G = (V, E)$ with nodes $0, 1, \ldots, n-1$ and edge weights w_{ij} ($w_{ij} = 1$ unless stated); $W = \sum_{(i,j)\in E} w_{ij}$. A split is a bitstring $z = z_0 z_1 \ldots z_{n-1}$ ($z_i \in \{0, 1\}$ = group of node i, **node 0 written first**); $\bar{z}$ is z with every bit flipped. Spins: $s_i = (-1)^{z_i} = 1 - 2z_i$, the eigenvalue of Z_i on $|z\rangle$.
- Cut value $C(z) = \sum_{(i,j)\in E} w_{ij}\,[z_i \neq z_j] = \sum_{(i,j)} w_{ij} \frac{1-s_i s_j}{2}$; $C_{\max} = \max_z C(z)$.
- **Notebook convention** (`QAOA.ipynb`): $H_C = \sum_{(i,j)\in E} \frac{w_{ij}}{2} Z_i Z_j$, so $C = \frac{W}{2} - H_C$; $H_M = \sum_i X_i$; $|\boldsymbol{\gamma}, \boldsymbol{\beta}\rangle = e^{-i\beta_p H_M} e^{-i\gamma_p H_C} \cdots e^{-i\beta_1 H_M} e^{-i\gamma_1 H_C} |+\rangle^{\otimes n}$; $F_p(\boldsymbol{\gamma}, \boldsymbol{\beta}) = \langle \boldsymbol{\gamma}, \boldsymbol{\beta}| C |\boldsymbol{\gamma}, \boldsymbol{\beta}\rangle$; approximation ratio $r = F_p / C_{\max}$. The angle vector is `params` $= [\gamma_1, \ldots, \gamma_p, \beta_1, \ldots, \beta_p]$.
- Gates: $H = \frac{1}{\sqrt{2}} \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$; X, Y, Z are the Pauli matrices; $R_X(\theta) = e^{-i\theta X/2}$, $R_Z(\theta) = e^{-i\theta Z/2}$; CNOT$(c, t)$ flips qubit t when qubit c is 1. Two-qubit matrices use the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ (lower-numbered qubit first); $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$.
- **Notebook defaults:** random graph with 15 nodes and edge probability 0.3 (`SEED = 42`: 30 edges), $p = 2$, COBYLA with 80 evaluations, ramp initial angles, 4000 shots, exact objective via `cudaq.observe`. A *previously recorded run* means the outputs stored in the notebook (CUDA-Q 0.16.0, CPU target `qpp-cpu`).
- All measurements are in the computational basis; ln is the natural logarithm. Numerical answers are rounded.

SECTION A · Mathematics and Graph Theory for QAOA Q.1–20

Syllabus: graphs, cuts and Max-Cut, bipartite graphs and odd cycles, complexity, QUBO and Ising forms, binary and spin variables, probability and sampling statistics, matrix exponentials and commutators.
Formats: Single-correct MCQ (Q.1–19); Matching lists (Q.2–13); Assertion–Reason (Q.7–20); Statement I / II (Q.8–11); Paragraph based (Q.17–18). Each question: +3 / 0 / −1.

Q.1 A graph has $n = 15$ nodes. The number of bitstrings $z \in \{0,1\}^{15}$, and the number of *distinct* partitions $\{S, V \setminus S\}$ of the node set (unordered; $S = \varnothing$ allowed) that these bitstrings describe, are

(A) 32 768 and 16 384 **(B)** 16 384 and 8 192 **(C)** 32 768 and 16 383 **(D)** 32 768 and 32 768

Q.2 Match each graph (List-I) with its maximum cut $C_{\max}$ (List-II). All edges have weight 1.

List-I		List-II	
(P)	cycle C_9 (9 nodes, 9 edges)	**(1)**	12
(Q)	complete graph K_5	**(2)**	9
(R)	complete bipartite graph $K_{3,3}$	**(3)**	6
(S)	Petersen graph (10 nodes, 15 edges, 3-regular)	**(4)**	8

(A) P→4, Q→3, R→2, S→1 **(B)** P→2, Q→3, R→1, S→4
(C) P→4, Q→2, R→3, S→1 **(D)** P→4, Q→3, R→1, S→2

Q.3 A weighted graph has the edges $(0,1)$ with weight 3, $(1,2)$ with weight 1, $(2,3)$ with weight 2, $(3,0)$ with weight 4 and $(0,2)$ with weight 2. Every node is put into group 0 or group 1 independently with probability $\frac{1}{2}$. The expected cut value is

(A) 12 **(B)** 6 **(C)** 3 **(D)** 2.4

Q.4 For the path graph $0-1-2$ (two unit edges), the cut value written as a polynomial in the binary variables $z_0, z_1, z_2 \in \{0,1\}$ is

(A) $z_0 + 2z_1 + z_2 - z_0z_1 - z_1z_2$ **(B)** $z_0 + 2z_1 + z_2 - 2z_0z_1 - 2z_1z_2$
(C) $z_0 + z_1 + z_2 - 2z_0z_1 - 2z_1z_2$ **(D)** $2 - z_0 - 2z_1 - z_2 + 2z_0z_1 + 2z_1z_2$

Q.5 The notebook's default graph is a random graph on $n = 15$ nodes in which each of the possible edges is present independently with probability 0.3. The expected number of edges and its standard deviation are

(A) 45 and 3.97 **(B)** 31.5 and 22.05 **(C)** 63 and 6.64 **(D)** 31.5 and 4.70

Q.6 The 15-node graph of the notebook has the degree sequence (nodes 0 to 14)

$$(d_0, \ldots, d_{14}) = (2, 3, 5, 4, 5, 4, 6, 1, 4, 3, 4, 2, 4, 6, 7).$$

The number of edges, and the number of nodes of odd degree, are

(A) 30 and 5 **(B)** 60 and 6 **(C)** 30 and 6 **(D)** 30 and 9

Q.7 **Assertion (A):** The cycle C_7 (7 nodes, 7 unit edges) has $C_{\max} = 6$.
Reason (R): Every graph with an odd number of nodes has $C_{\max} < |E|$.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Q.8 **Statement I:** A polynomial-time algorithm that solves Max-Cut *exactly* on every graph is known.
Statement II: Finding a *minimum* cut of a graph (both groups non-empty, non-negative weights) is NP-hard.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Q.9 For the triangle K_3 (unit weights) consider $H_C = \frac{1}{2}(Z_0Z_1 + Z_1Z_2 + Z_0Z_2)$. Its smallest eigenvalue and the number of computational basis states with that eigenvalue are

(A) $-\frac{1}{2}$, six basis states **(B)** $-\frac{1}{2}$, three basis states

(C) $+\frac{3}{2}$, two basis states **(D)** $-\frac{3}{2}$, two basis states

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Q.10 For the notebook's 15-node graph (30 unit edges), a split z is chosen uniformly at random from all 2^{15} bitstrings. The variance of the cut value $C(z)$ is

(A) 15 **(B)** 3.75 **(C)** 7.5 **(D)** 30

Q.11 **Statement I:** Because all the operators Z_iZ_j commute with one another,
$e^{-i\gamma H_C} = \prod_{(i,j)\in E} e^{-i\gamma \frac{w_{ij}}{2} Z_iZ_j}$ holds exactly, with the factors in any order.
Statement II: Because the operators X_i also commute with one another,
$e^{-i\beta H_M}\, e^{-i\gamma H_C} = e^{-i(\beta H_M + \gamma H_C)}$ holds for all β, γ.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Q.12 The expectation value $\langle Z_iZ_j \rangle$ is estimated as the average of the ± 1 values $(-1)^{z_i \oplus z_j}$ over S independent shots. The smallest S that guarantees a standard error of at most 0.01 *for every state* is

(A) 10 000 **(B)** 2 500 **(C)** 100 **(D)** 40 000

Q.13 In the previously recorded notebook run, an optimal split is sampled with probability $P_{\text{opt}} = 0.06$ per shot. Treat the shots as independent and match each quantity (List-I) with its value (List-II).

List-I		**List-II**	
(P)	standard deviation of the number of optimal shots among 4000	**(1)**	75
(Q)	probability that 10 shots contain *no* optimal split	**(2)**	15.0
(R)	probability that 10 shots contain *at least one* optimal split	**(3)**	0.539
(S)	smallest number of shots giving $\geq 99\,\%$ probability of at least one optimal split	**(4)**	0.461

(A) P→2, Q→3, R→4, S→1 **(B)** P→2, Q→4, R→3, S→1
(C) P→1, Q→4, R→3, S→2 **(D)** P→1, Q→3, R→4, S→2

Q.14 The notebook's brute-force search reports four optimal bitstrings (cut value 25):

`001101110100010`, `001101110110010`, `110010001001101`, `110010001011101`.

The number of distinct optimal partitions, and the bitstring describing the same partition as `001101110100010`, are

(A) 2; `110010001001101` **(B)** 2; `010001011101100`
(C) 4; `110010001011101` **(D)** 2; `110010001011101`

Q.15 The QUBO function $f(x_0, x_1) = 2x_0x_1 - x_0 - x_1$ with $x_i \in \{0, 1\}$ is rewritten with spins via $x_i = (1 - s_i)/2$, $s_i \in \{+1, -1\}$. The result is

(A) $\frac{1}{2}\, s_0 s_1 - 1 + \frac{1}{2}(s_0 + s_1)$ **(B)** $\frac{1}{2}\,(s_0 s_1 - 1)$
(C) $\frac{1}{4}\,(s_0 s_1 - 1)$ **(D)** $\frac{1}{2}\,(1 - s_0 s_1)$

Q.16 Let $f(\theta) = a + b\cos\theta + c\sin\theta$ with constants a, b, c. Which expression equals $f'(\theta)$ *exactly* for every θ?

(A) $\frac{1}{2}\big[f(\theta + \frac{\pi}{4}) - f(\theta - \frac{\pi}{4})\big]$ **(B)** $f(\theta + \frac{\pi}{2}) - f(\theta)$
(C) $\frac{1}{2}\big[f(\theta + \frac{\pi}{2}) - f(\theta - \frac{\pi}{2})\big]$ **(D)** $\frac{1}{\pi}\big[f(\theta + \frac{\pi}{2}) - f(\theta - \frac{\pi}{2})\big]$

Paragraph for Questions 17 and 18

The "house graph" has the 5 nodes $0, 1, 2, 3, 4$ and the 6 unit-weight edges $(0,1), (1,2), (2,3), (3,0)$ (a square) and $(0,4), (1,4)$ (a roof: node 4 forms a triangle with nodes 0 and 1). A split is a bitstring $z = z_0 z_1 z_2 z_3 z_4$; its cut value $C(z)$ counts the edges whose two end nodes are in different groups.

Q.17 The maximum cut $C_{\max}$ of the house graph, and the number of bitstrings z that achieve it, are

(A) 5 and 6 **(B)** 5 and 2 **(C)** 6 and 2 **(D)** 5 and 4

Q.18 A split of the house graph is drawn uniformly at random. The probability that it is optimal, and the ratio $\mathbb{E}[C]/C_{\max}$ of the random split, are

(A) $\frac{1}{16}$ and 0.6 **(B)** $\frac{1}{4}$ and 0.6 **(C)** $\frac{1}{8}$ and 0.5 **(D)** $\frac{1}{8}$ and 0.6

Q.19 For the Petersen graph (10 nodes, 15 unit edges, $C_{\max} = 12$), the smallest eigenvalue of $H_C = \sum_{(i,j)\in E} \frac{1}{2} Z_i Z_j$ is

(A) -7.5 **(B)** -4.5 **(C)** -12 **(D)** $+4.5$

Q.20 The notebook's `local_search_maxcut` repeatedly flips single nodes while this strictly increases the cut (unit weights), starting from random splits.

Assertion (A): Every split returned by one run of this greedy procedure is an optimal split.

Reason (R): When no single flip increases the cut, every node has at least half of its incident edges cut.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

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SECTION B · Quantum Mechanics Q.21–40

Syllabus: qubits, superposition and the Born rule, Pauli algebra, rotation gates, CNOT conjugation, expectation values, entanglement, global phases, symmetries and state-vector memory.
Formats: Single-correct MCQ (Q.21–37); Matching lists (Q.23–27); Assertion–Reason (Q.35); Statement I / II (Q.29–38); Paragraph based (Q.39–40). Each question: +3 / 0 / −1.

Q.21 The 15-qubit state $|+\rangle^{\otimes 15}$ is measured in the computational basis. The notebook's graph has 4 optimal bitstrings. The probability of obtaining an optimal bitstring, and the factor by which the previously recorded QAOA value $P_{\text{opt}} = 0.06$ exceeds it, are

(A) 6.10×10^{-5}; about 980 **(B)** 3.05×10^{-5}; about 1970
(C) 2.44×10^{-4}; about 245 **(D)** 1.22×10^{-4}; about 490

Q.22 The two-qubit operator $(Z \otimes Z)(X \otimes X)$ equals

(A) $i\,Y \otimes Y$ **(B)** $-\,Y \otimes Y$ **(C)** $Y \otimes Y$ **(D)** $-i\,Y \otimes Y$

Q.23 Match each operator (List-I) with its matrix (List-II). Two-qubit matrices use the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

List-I		List-II	
(P)	$R_X(2\beta)$	**(1)**	$\text{diag}\,(e^{-i\theta/2}, e^{i\theta/2}, e^{i\theta/2}, e^{-i\theta/2})$
(Q)	$R_Z(\theta)$	**(2)**	$\begin{pmatrix} \cos\beta & -i\sin\beta \\ -i\sin\beta & \cos\beta \end{pmatrix}$
(R)	$e^{-i\theta\, Z\otimes Z/2}$	**(3)**	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$
(S)	H	**(4)**	$\text{diag}\,(e^{-i\theta/2}, e^{i\theta/2})$

(A) P→3, Q→4, R→1, S→2 **(B)** P→2, Q→1, R→4, S→3
(C) P→4, Q→2, R→1, S→3 **(D)** P→2, Q→4, R→1, S→3

Q.24 The state $R_Z(\theta)\,|+\rangle$ is passed through a Hadamard gate and then measured in the computational basis. The probability of the outcome 0 is

(A) $\cos^2(\theta/2)$ **(B)** $\sin^2(\theta/2)$ **(C)** $\cos^2\theta$ **(D)** $\frac{1}{2}$

Q.25 For one unit edge $(0, 1)$ the qubits are in the state $|\psi\rangle = \frac{1}{2}|00\rangle + \frac{\sqrt{3}}{2}|01\rangle$. The expected cut value $\langle C\rangle = \langle \frac{1}{2}(1 - Z_0 Z_1)\rangle$ is

(A) $\frac{1}{4}$ **(B)** $-\frac{1}{2}$ **(C)** $\frac{\sqrt{3}}{2}$ **(D)** $\frac{3}{4}$

Q.26 Which of the following operators commutes with $Z_0 Z_1$ (on two or more qubits)?

(A) $X_0 X_1$ **(B)** $Y_0 Z_1$ **(C)** X_0 **(D)** X_1

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Q.27 Conjugation by a CNOT with control qubit 0 and target qubit 1 maps a Pauli operator P to CNOT P CNOT. Match each P (List-I) with its image (List-II).

List-I		**List-II**	
(P)	$X \otimes I$	**(1)**	$Z \otimes Z$
(Q)	$I \otimes X$	**(2)**	$X \otimes X$
(R)	$Z \otimes I$	**(3)**	$I \otimes X$
(S)	$I \otimes Z$	**(4)**	$Z \otimes I$

(A) P→3, Q→2, R→1, S→4 **(B)** P→2, Q→3, R→1, S→4
(C) P→3, Q→2, R→4, S→1 **(D)** P→2, Q→3, R→4, S→1

Q.28 The state $e^{-i\gamma\, Z\otimes Z/2}\,|+\rangle\,|+\rangle$ is maximally entangled for which $\gamma \in (0, \pi)$?

(A) $\gamma = \pi/2$ **(B)** $\gamma = 3\pi/4$
(C) $\gamma = \pi/4$ **(D)** for no γ: it is always a product state

Q.29 **Statement I:** Replacing H_C by $H_C + c\,I$ (any real constant c) changes the QAOA state $|\boldsymbol{\gamma}, \boldsymbol{\beta}\rangle$ only by a global phase.

Statement II: Consequently, a QAOA circuit that uses $e^{-i\gamma C}$ with $C = \frac{W}{2} - H_C$ has the same optimal angles (γ^*, β^*) as the notebook's circuit with $e^{-i\gamma H_C}$.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Q.30 Two qubits are in a product state in which $P(z_0 = 1) = 0.2$ and $P(z_1 = 1) = 0.7$. The value of $\langle Z_0 Z_1 \rangle$ is

(A) $+0.24$ **(B)** -0.24 **(C)** 0.14 **(D)** 0.62

Q.31 Conjugating Z by the mixer gate gives $e^{i\beta X}\, Z\, e^{-i\beta X} =$

(A) $\cos 2\beta\; Z - \sin 2\beta\; Y$ **(B)** $\cos 2\beta\; Z + \sin 2\beta\; Y$ **(C)** $\cos\beta\; Z + \sin\beta\; Y$ **(D)** $\cos 2\beta\; Z + \sin 2\beta\; X$

Q.32 A state vector needs 16×2^n bytes in fp64 and 8×2^n bytes in fp32. Ignoring simulator workspace, the largest n whose state fits on a single 80 GB (80×10^9 bytes) GPU is

(A) 33 (fp64) and 34 (fp32) **(B)** 36 (fp64) and 37 (fp32)
(C) 32 (fp64) and 33 (fp32) **(D)** 32 (fp64) and 64 (fp32)

Q.33 The gate $R_X(2\beta)$ with $\beta = \pi/8$ is applied to $|0\rangle$. The probability of measuring 1 is

(A) $\sin(\pi/8) \approx 0.383$ **(B)** $\frac{1}{2}$
(C) $\frac{2-\sqrt{2}}{4} \approx 0.146$ **(D)** $\frac{2+\sqrt{2}}{4} \approx 0.854$

Q.34 Which statement about $|+\rangle^{\otimes n}$ and $H_M = \sum_i X_i$ is correct?

(A) It is an eigenstate of H_M with eigenvalue 0

(B) It is the eigenstate of H_M with the *largest* eigenvalue, $+n$

(C) It is not an eigenstate of H_M

(D) It is the ground state of H_M, with eigenvalue $-n$

Q.35 **Assertion (A):** A state $|\psi\rangle$ and its complex conjugate $|\psi^*\rangle$ (every amplitude conjugated) give identical computational-basis measurement statistics.

Reason (R): The initial QAOA state $|+\rangle^{\otimes n}$ has only real amplitudes.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Q.36 For the triangle K_3 (unit weights) a state has $\langle Z_0Z_1\rangle = -0.6$, $\langle Z_1Z_2\rangle = -0.2$ and $\langle Z_0Z_2\rangle = 0.4$. Its expected cut $\langle C\rangle$ is

(A) -0.2 **(B)** 1.9 **(C)** 1.3 **(D)** 1.7

Q.37 On n qubits, $e^{-i(\pi/2)H_M}$ with $H_M = \sum_i X_i$ equals ..., and therefore (since $C(z) = C(\bar{z})$) F_p is periodic in every single β_l with period ...

(A) $(-i)^n X^{\otimes n}$; $\pi/2$ **(B)** $(-i)^n X^{\otimes n}$; π

(C) I; $\pi/2$ **(D)** $X^{\otimes n}$; 2π

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Q.38 **Statement I:** Z_iZ_j has eigenvalue -1 exactly on the computational basis states with $z_i \neq z_j$.

Statement II: $\langle Z_iZ_j\rangle$ can be estimated from the same computational-basis shots that are used to sample candidate splits.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Paragraph for Questions 39 and 40

A single edge $(0, 1)$ with weight 1 is treated with one QAOA layer in the notebook's convention. Starting from $|00\rangle$ the gates are: H on both qubits; CNOT (control 0, target 1); $R_Z(\theta)$ on qubit 1 with $\theta = \gamma w = \gamma$; CNOT (control 0, target 1); $R_X(2\beta)$ on both qubits. The angles are $\gamma = \pi/2$ and $\beta = \pi/8$. Amplitudes are listed in the order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ (qubit 0 written first).

Q.39 Immediately after the second CNOT (before the mixer), the phase of the amplitude of $|01\rangle$ relative to the amplitude of $|00\rangle$ is

(A) $45°$ **(B)** $180°$ **(C)** $90°$ **(D)** $0°$

Q.40 After the mixer, the probability of measuring a *cut* bitstring (01 or 10) is

(A) 1 **(B)** $\frac{3}{4}$ **(C)** 0 **(D)** $\frac{1}{2}$

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SECTION C · QAOA Basics Q.41–60

Syllabus: cost and mixer Hamiltonians, initial state, QAOA circuit and gate decomposition, parameters and depth, the $p = 1$ analysis, objective function, sign conventions and the notebook's code.
Formats: Single-correct MCQ (Q.41–57); Matching lists (Q.46–55); Assertion–Reason (Q.53); Statement I / II (Q.49–58); Paragraph based (Q.59–60). Each question: +3 / 0 / −1.

Q.41 For `edges = [(0, 1, 2.0), (1, 2, 1.0)]` the notebook's function `cost_hamiltonian()` returns the operator

(A) $1.0\,Z_0Z_1 + 0.5\,Z_1Z_2$ **(B)** $-1.0\,Z_0Z_1 - 0.5\,Z_1Z_2$
(C) $2.0\,Z_0Z_1 + 1.0\,Z_1Z_2$ **(D)** $1.5\,I - 1.0\,Z_0Z_1 - 0.5\,Z_1Z_2$

Q.42 In the notebook's convention, if all cost angles are zero ($\gamma_1 = \cdots = \gamma_p = 0$), the expected cut $F_p(\boldsymbol{\gamma}, \boldsymbol{\beta})$ equals

(A) $W/2 + \frac{1}{2}\sum_l \sin 4\beta_l$ **(B)** $C_{\max}$
(C) $W/2$ for every choice of $\boldsymbol{\beta}$ **(D)** 0

Q.43 In the notebook's kernel each edge (i, j) with weight w is implemented as `x.ctrl(q[i], q[j]); rz(gamma * w, q[j]); x.ctrl(q[i], q[j])`. With $R_Z(\theta) = e^{-i\theta Z/2}$, this sequence equals

(A) $e^{+i\gamma\,(w/2)\,Z_iZ_j}$ **(B)** $e^{-i\gamma\,w\,Z_iZ_j}$ **(C)** $e^{-i\gamma\,(w/2)\,Z_j}$ **(D)** $e^{-i\gamma\,(w/2)\,Z_iZ_j}$

Q.44 Why does the notebook apply `rx(2.0 * beta, q)` (and not `rx(beta, q)`) in the mixer layer?

(A) Because the mixer must be applied twice per layer
(B) Because the mixer Hamiltonian is $H_M = 2\sum_i X_i$
(C) Because β is measured in units of $\pi/2$
(D) Because $R_X(\theta) = e^{-i\theta X/2}$, so $R_X(2\beta) = e^{-i\beta X}$, the factor of $e^{-i\beta H_M}$ for one qubit

Q.45 Using the notebook's gate structure (H on every qubit, then per layer 3 gates per edge and one R_X per qubit), a graph with $n = 20$ nodes and $|E| = 57$ edges at depth $p = 3$ needs how many gates (excluding measurements), and how many of them are CNOTs?

(A) 573 and 342 **(B)** 593 and 342 **(C)** 593 and 171 **(D)** 402 and 228

Q.46 For the notebook's default run ($n = 15$ nodes, $|E| = 30$ edges, $p = 2$), match each quantity (List-I) with its value (List-II).

List-I		List-II	
(P)	number of variational parameters	**(1)**	105
(Q)	number of R_Z gates	**(2)**	120
(R)	number of single-qubit gates (H, R_Z and R_X together)	**(3)**	4
(S)	number of CNOT (`cx`) gates	**(4)**	60

(A) P→4, Q→3, R→1, S→2 **(B)** P→3, Q→4, R→2, S→1
(C) P→3, Q→4, R→1, S→2 **(D)** P→3, Q→2, R→1, S→4

Q.47 For a single unit edge and $p = 1$ in the notebook's convention, the expected cut is $F_1(\gamma, \beta) = \frac{1}{2} - \frac{1}{2}\sin 4\beta \, \sin \gamma$. Its maximum value, and one point where it is reached, are

(A) $\frac{3}{4}$, at $(\gamma, \beta) = (\pi/2, -\pi/8)$ **(B)** $\frac{1}{2}$, reached for all angles
(C) 1, at $(\gamma, \beta) = (\pi/2, \pi/8)$ **(D)** 1, at $(\gamma, \beta) = (\pi/2, -\pi/8)$

Q.48 With optimally chosen angles, the best approximation ratio of $p = 1$ QAOA on the ring (cycle) C_8 is

(A) 1 **(B)** 3/4 **(C)** 5/6 **(D)** 0.6924

Q.49 **Statement I:** For $p = 1$ with optimal angles, QAOA achieves $\langle C \rangle \geq 0.6924\, C_{\max}$ on every 3-regular graph.

Statement II: For every graph, the optimal value $F_p^* = \max_{\boldsymbol{\gamma},\boldsymbol{\beta}} F_p$ satisfies $F_{p+1}^* \geq F_p^*$.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Q.50 Which statement about the limit of large depth p (angles optimised for each p) is correct?

(A) $F_p^*/C_{\max} \to 0.878$, the Goemans–Williamson ratio
(B) $F_p^* = C_{\max}$ already at $p = 1$ for every graph
(C) $F_p^* \to W/2$, because deep circuits randomise the state
(D) $F_p^* \to C_{\max}$ as $p \to \infty$, because a Trotterised adiabatic path is one admissible choice of angles

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Q.51 In the notebook the angle vector is `params` $= [\gamma_1, \ldots, \gamma_p, \beta_1, \ldots, \beta_p]$; the kernel reads `gamma = params[layer]` and `beta = params[layers + layer]`. For $p = 3$ and `params = [a, b, c, d, e, f]`, the mixer angle β_2 of the second layer is

(A) d **(B)** b **(C)** e **(D)** f

Q.52 With `INITIAL_PARAMETERS = "ramp"` the notebook starts from `np.linspace(0.2, 0.6, p)` for the γ's and `np.linspace(0.6, 0.2, p)` for the β's. For $p = 5$, the initial values of γ_3 and β_2 are

(A) 0.3 and 0.5 **(B)** 0.36 and 0.52 **(C)** 0.4 and 0.3 **(D)** 0.4 and 0.5

Q.53 **Assertion (A):** The notebook's objective function returns $\langle H_C \rangle$ and SciPy *minimises* it; this is equivalent to *maximising* the expected cut.

Reason (R): $\langle C \rangle = W/2 - \langle H_C \rangle$, and $W/2$ does not depend on the angles.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

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Q.54 At the notebook's default starting point $\boldsymbol{\gamma} = (0.2, 0.6)$, $\boldsymbol{\beta} = (0.6, 0.2)$, an exact state-vector calculation for its 15-node graph gives $\langle C \rangle \approx 9.18$, *below* the random baseline $W/2 = 15$. The best explanation is

(A) In the convention $e^{-i\beta H_M} e^{-i\gamma H_C}$, the start state $|+\rangle^{\otimes n}$ is the *highest* eigenstate of H_M, so an annealing-like ramp with $\gamma, \beta > 0$ heads for the highest eigenstate of H_C, the *smallest* cut; heading for the largest cut needs γ and β of opposite sign

(B) The value is meaningless: `cudaq.observe` returns random numbers before the optimizer has converged

(C) Every $p = 2$ starting point gives $\langle C \rangle < W/2$; only $p \geq 3$ can beat random guessing

(D) It indicates a bug, because a QAOA state can never have $\langle C \rangle < W/2$

Q.55 Match each part of the QAOA algorithm (List-I) with the code that implements it in the notebook (List-II).

List-I		List-II	
(P)	initial state $\|+\rangle^{\otimes n}$	**(1)**	`rx(2.0 * beta, qubits[q])`
(Q)	cost unitary for one edge	**(2)**	`cudaq.observe(kernels["state"], H_C, ...).expectation()`
(R)	mixer unitary for one qubit	**(3)**	`h(qubits)`
(S)	objective $\langle H_C \rangle$	**(4)**	`x.ctrl(...); rz(gamma * weights[e], ...); x.ctrl(...)`

(A) P→3, Q→1, R→4, S→2 **(B)** P→3, Q→4, R→2, S→1

(C) P→3, Q→4, R→1, S→2 **(D)** P→4, Q→3, R→1, S→2

Q.56 For the triangle K_3 (unit weights, $C_{\max} = 2$), the best value of F_1 over all angles is

(A) $\sqrt{3}$, i.e. $r \approx 0.866$ **(B)** 2, i.e. $r = 1$

(C) 1.75, i.e. $r = 0.875$ **(D)** 1.5, i.e. $r = 0.75$

Q.57 The previously recorded run reports: approximation ratio $r = \langle C \rangle / C_{\max} = 0.8213$, best sampled cut 25 ($= C_{\max}$), and $P(\text{optimal split}) = 0.06$ with 4000 shots. Which interpretation is correct?

(A) 82 % of the shots returned an optimal split

(B) r measures the *average* cut of the distribution; the best of 4000 samples can still be optimal, and here 240 shots were optimal

(C) The best sampled cut can never exceed $\langle C \rangle = 20.53$, so the value 25 indicates an error

(D) The best sampled split has cut $0.8213 \times 25 \approx 20.5$

Q.58 **Statement I:** The notebook's final sampling run uses the angles of the best evaluation stored in `history_cut`, which need not be the optimizer's final point `opt.x`.

Statement II: With COBYLA and a budget of 80 evaluations, the notebook is guaranteed to find the global maximum of F_2.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Paragraph for Questions 59 and 60

QAOA with $p = 1$ is applied to the path graph $0 - 1 - 2$ (edges $(0, 1)$ and $(1, 2)$, unit weights, $C_{\max} = 2$) in the notebook's convention, with $\gamma = \pi/2$ and $\beta = -\pi/8$. For a triangle-free graph and $p = 1$ the edge correlator in this convention is

$$\langle Z_u Z_v \rangle = \tfrac{1}{2} \sin 4\beta \, \sin \gamma \, \big(\cos^{d_u - 1} \gamma + \cos^{d_v - 1} \gamma \big),$$

where d_u, d_v are the degrees of the two end nodes.

Q.59 The value of $\langle Z_0 Z_1 \rangle$ is

(A) $-\frac{1}{2}$ **(B)** -1 **(C)** $+\frac{1}{2}$ **(D)** 0

Q.60 The expected cut $\langle C \rangle$ and the probability of sampling an optimal split are

(A) 1.5 and 0.3125 **(B)** 1.5 and 0.625 **(C)** 1.25 and 0.625 **(D)** 2 and 1

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SECTION D · Advanced QAOA and Implementation Q.61–80

Syllabus: light cones, gradients and classical optimizers, shot noise, parameter strategies, simulation backends and memory, noise, QAOA variants, limitations, benchmarking and interpretation of recorded results.
Formats: Single-correct MCQ (Q.61–78); Matching lists (Q.68–72); Assertion–Reason (Q.70); Statement I / II (Q.74); Paragraph based (Q.79–80). Each question: +3 / 0 / −1.

Q.61 On a ring graph, the expectation $\langle Z_u Z_v \rangle$ for one edge (u, v) of a depth-p QAOA state depends on at most how many qubits when $p = 3$?

(A) 8 **(B)** all n qubits **(C)** 7 **(D)** 6

Q.62 For even rings C_n with $n \geq 2p + 2$, numerical optimisation of the angles gives the best $p = 2$ approximation ratio

(A) 3/4 **(B)** 5/6 **(C)** 7/8 **(D)** 1

Q.63 For the default circuit ($n = 15$, $|E| = 30$, $p = 2$), compare two ways of computing the gradient of F_2 with respect to all $2p$ angles: (i) the exact parameter-shift rule applied to every rotation gate separately (each R_Z and R_X gate is shifted by $\pm\pi/2$ on its own), and (ii) a forward finite-difference gradient as used by L-BFGS-B. The numbers of circuit evaluations are

(A) 8 and 5 **(B)** 180 and 4 **(C)** 90 and 5 **(D)** 180 and 5

Q.64 The notebook optimises the angles with SciPy's `minimize(..., method="COBYLA")` and the options `{"maxiter": 80, "rhobeg": 0.2}`. The option `rhobeg = 0.2` sets

(A) a learning rate multiplying the gradient
(B) the maximum number of iterations divided by 400
(C) the tolerance at which COBYLA declares convergence
(D) the initial size of the trust region, i.e. the first steps change the angles by about 0.2 rad

Q.65 With `OBJECTIVE_SHOTS = 1000`, $\langle H_C \rangle$ is estimated from 1000 shots. If the cut values of the sampled bitstrings have a standard deviation of 2.5, the standard error of the estimated objective is about

(A) 0.0063 **(B)** 0.0025 **(C)** 0.079 **(D)** 2.5

Q.66 For $n = 22$ (`BRUTE_FORCE_MAX_NODES`), the notebook's `brute_force_maxcut` creates the arrays `index = np.arange(2**N, dtype=np.int64)` and `values = np.zeros(2**N)`. Together they occupy about

(A) 34 MB **(B)** 67 MB **(C)** 0.5 MB **(D)** 4.2 MB

Q.67 A 36-qubit QAOA state vector is simulated in fp64. If it is split evenly over a power-of-two number of 80 GB GPUs (as the multi-GPU state-vector target does), ignoring workspace, the minimum number of GPUs is

(A) 8 **(B)** 14 **(C)** 32 **(D)** 16

Q.68 Match each setting of `BACKEND` and `SIMULATION_METHOD` in `QAOA.ipynb` (List-I) with what the notebook's `resolve_backend` does (List-II).

List-I		List-II	
(P)	`"cpu"` + `"statevector"`	**(1)**	target `tensornet-mps`
(Q)	`"cpu"` + `"density_matrix"`	**(2)**	stops with an error: the MPS simulator needs an NVIDIA GPU
(R)	`"gpu"` + `"mps"`	**(3)**	target `qpp-cpu`
(S)	`"cpu"` + `"mps"`	**(4)**	target `density-matrix-cpu`

(A) P→4, Q→3, R→1, S→2 **(B)** P→3, Q→1, R→4, S→2
(C) P→3, Q→4, R→2, S→1 **(D)** P→3, Q→4, R→1, S→2

Q.69 For which instance is a matrix-product-state (MPS) simulation of a depth-p QAOA circuit most likely to be efficient (small bond dimension), with the qubits ordered along a line $0, 1, \ldots, n-1$?

(A) Any graph, because QAOA states are always product states before measurement
(B) A complete graph K_n: its symmetry makes the MPS exact with bond dimension 1
(C) A ring C_n with small p: every cut of the line is crossed by only 2 edges
(D) A dense random graph with edge probability 0.5: more edges average out the entanglement

Q.70 **Assertion (A):** When `BACKEND = "hardware"`, the notebook estimates the objective from shots even if `OBJECTIVE_SHOTS = None`.
Reason (R): Hardware (and hardware-emulation) targets only return measurement samples; they cannot give a state vector or an exact expectation value.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

Q.71 A simple global-depolarising noise model treats the noisy QAOA output as a mixture: with probability $F = (1-\varepsilon)^{N_{\mathrm{CNOT}}}$ the ideal state, otherwise a uniformly random bitstring. For the default circuit ($N_{\mathrm{CNOT}} = 120$, ideal $\langle C \rangle = 20.53$, $W/2 = 15$) and $\varepsilon = 0.005$ per CNOT, the predicted noisy $\langle C \rangle$ is about

(A) 17.2 **(B)** 15.0 **(C)** 18.0 **(D)** 11.3

Q.72 Match each QAOA variant (List-I) with its key idea (List-II).

List-I		List-II	
(P)	warm-start QAOA	**(1)**	problem-specific mixers (e.g. XY mixers) that keep the state inside the feasible subspace of a constrained problem
(Q)	ADAPT-QAOA	**(2)**	use measured correlations $\langle Z_i Z_j \rangle$ to fix and eliminate variables, then solve the smaller instance
(R)	Quantum Alternating Operator Ansatz (Hadfield et al.)	**(3)**	start from a state biased towards a classical (e.g. relaxed/SDP) solution instead of $\lvert+\rangle^{\otimes n}$
(S)	recursive QAOA (RQAOA)	**(4)**	grow the mixer layer by layer from an operator pool, choosing the operator with the largest energy gradient

(A) P→3, Q→1, R→4, S→2 **(B)** P→3, Q→4, R→2, S→1
(C) P→3, Q→4, R→1, S→2 **(D)** P→4, Q→3, R→1, S→2

Q.73 The INTERP initialisation (Zhou et al. 2020) builds depth-$(p+1)$ starting angles from optimised depth-p angles by $\gamma_i^{(p+1)} = \frac{i-1}{p}\,\gamma_{i-1}^{(p)} + \frac{p-i+1}{p}\,\gamma_i^{(p)}$ for $i = 1, \ldots, p+1$ (with $\gamma_0^{(p)} = \gamma_{p+1}^{(p)} = 0$). From $\boldsymbol{\gamma}^{(2)} = (0.4, 0.8)$ it gives $\boldsymbol{\gamma}^{(3)} =$

(A) $(0.4,\ 0.6,\ 0.8)$ **(B)** $(0.27,\ 0.53,\ 0.8)$
(C) $(0.4,\ 0.8,\ 0.8)$ **(D)** $(0.4,\ 0.8,\ 0)$

Q.74 For graphs with more nodes than `BRUTE_FORCE_MAX_NODES`, the notebook replaces the exact optimum by the best cut found by `local_search_maxcut` and still reports $r = \langle C \rangle / $`max_cut`.

Statement I: In that case the reported `max_cut` is guaranteed to be at least the true $C_{\max}$.

Statement II: In that case the reported ratio is at least the true ratio $\langle C \rangle / C_{\max}$.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Q.75 In the recorded run, the quantum execution time was 40.01 s for the 80 objective evaluations (`cudaq.observe` calls) of the 225-gate circuit on 15 qubits. The average time per evaluation and per gate are about

(A) 0.010 s and 44 μs **(B)** 0.18 s and 0.8 ms
(C) 0.50 s and 2.2 ms **(D)** 40 s and 0.18 s

Q.76 The notebook estimates run time as (number of evaluations) × (gates per circuit) × (time per gate), with the time per gate scaling as 2^n for a state vector. Going from the default ($n = 15$, $|E| = 30$, $p = 2$) to $n = 25$ with $|E| = 90$ edges at the same p and evaluation budget, the run time grows by a factor of about

(A) 2.9×10^6 **(B)** 2.8×10^3 **(C)** 2.7 **(D)** 1.0×10^3

Q.77 Which statement about QAOA for Max-Cut reflects the current state of knowledge?

(A) No quantum advantage has been demonstrated: the classical Goemans–Williamson algorithm guarantees 0.878 in polynomial time, and fixed-depth QAOA is limited by locality

(B) The recorded result $r = 0.82$ on 15 nodes proves a quantum speed-up

(C) QAOA solves Max-Cut exactly in polynomial time for any fixed p

(D) QAOA with $p = 1$ beats Goemans–Williamson on every 3-regular graph

Q.78 With `GRAPH = "custom"` and `CUSTOM_EDGES = [(0, 1, 1.0), (1, 2, 2.5), (0, 2, -1.0)]` (3 nodes), the maximum cut and the random-split baseline printed by the notebook are

(A) 3.5 and 2.25 **(B)** 3.5 and 1.25 **(C)** 2.5 and 1.25 **(D)** 4.5 and 2.25

Paragraph for Questions 79 and 80

Previously recorded run of ***`QAOA.ipynb`*** (default settings, CUDA-Q 0.16.0, target `qpp-cpu`): 15 nodes, 30 unit edges ($W = 30$), exact optimum $C_{\max} = 25$ with the four optimal bitstrings listed in A14, $p = 2$, COBYLA with 80 evaluations, best expected cut $\langle C \rangle = 20.5336$, 4000 shots. The four most frequent bitstrings were all optimal: `110010001001101` (68 shots), `001101110110010` (67), `110010001011101` (56), `001101110100010` (49); in total 240 shots were optimal.

Q.79 Estimated from the recorded counts, the probability of sampling the partition described by `110010001001101` (either this bitstring or its complement) is

(A) 0.0338 **(B)** 0.0600 **(C)** 0.0170 **(D)** 0.0263

Q.80 A stricter figure of merit measures the improvement over random guessing: $\eta = \dfrac{\langle C \rangle - W/2}{C_{\max} - W/2}$. For the recorded run, η is

(A) 0.600 **(B)** 0.553 **(C)** 0.221 **(D)** 0.821

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